Lab Report I — A Step-by-Step Exemplar

Physics 2026–27 · seven sections, in order · built from your Day 1 and Day 2 sheets

How to use this. Write your report one step at a time. For each step, read the checklist, look at the exemplar, then write that section with **your group's numbers and your own words**. The exemplar uses a sample group's data; copying its numbers or sentences is not your report.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	STEP 7
Question	Hypothesis	Materials	Procedure	Data	Analysis	Conclusion

Step 1 · Question

ONE SENTENCE

- ☐ Write it as a question, ending in a question mark.
- ☐ Name both variables: what you changed (**time**) and what you measured (**position**).
- ☐ Say the conditions: a cart, starting from rest, on a ramp.

EXEMPLAR

Question

What is the relationship between position and time for a cart accelerating from rest down a ramp?

Step 2 · Hypothesis

FROM DAY 1, SECTION 2

- ☐ Use the form **If … then … because …**
- ☐ The *because* is physics: name the equation that backs your prediction.
- ☐ List what you can actually check with your data: the shape of each graph, and the time ratio from Day 1, 2(b).

EXEMPLAR

Hypothesis

If the cart's acceleration is constant, then its position will be proportional to the square of the time, because an object starting from rest with constant acceleration follows $x = \frac{1}{2}at^2$.

This predicts three things we can test:

- The position–time graph will curve upward, because the cart covers more distance in each second than in the one before.
- Reaching a mark four times as far from the release line will take twice as long, because $4 = 2^2$.
- A graph of position against time squared will be a straight line through the origin, with a slope equal to ½a.

Step 3 · Materials

A BULLETED LIST

- [ ] List everything you used, one item per bullet.
- [ ] Give sizes: how long your ramp was and how high you raised it (Day 1, section 1).
- [ ] For each measuring tool, give its precision, such as ±0.001 m or “displays to 0.01 s”.

EXEMPLAR

Materials

- Cardboard sheet, 1.20 m long, folded into a U-shaped channel so it does not sag
- Two textbooks, to raise the top end of the ramp 0.060 m
- Lab cart
- Meter stick (±0.001 m)
- Masking tape, for the release line, the position marks and the side rails
- Phone stopwatch (displays to 0.01 s)
- Day 1 and Day 2 lab sheets

Step 4 · Procedure

NUMBERED STEPS

- [ ] Number the steps. Another group should be able to repeat your experiment from them alone.
- [ ] Include how you built the ramp, where the origin was, which way was positive, and where the marks were.
- [ ] Say how a run was timed, how many trials per mark, and what you did with a bad trial.

EXEMPLAR

Procedure

1. Fold the cardboard into a channel 1.20 m long. Lay a meter stick inside it to check that it is straight with no sag.
2. Raise the top end 0.060 m with two textbooks. Tape both ends to the table and add a tape backstop at the bottom.

3. Test the slope: release the cart once and time the full length. It took about 2.4 s, which is longer than the 2 s minimum, so we kept that height.
4. Mark a release line near the top end. This is the origin, and down the ramp is the positive direction.
5. Measure from the release line and tape marks at 0.20, 0.40, 0.60, 0.80 and 1.00 m.
6. Place the front of the cart on the release line and let go without pushing. The timer starts the stopwatch on release and stops it when the front of the cart reaches the first mark.
7. Run three trials for each mark. Rotate the releaser, timer and recorder jobs after every mark. If a trial is clearly wrong, for example the cart hits the side rail, cross it out and rerun it.
8. Average the three times for each mark and plot position against average time.

Step 5 · Data

TWO TABLES + ONE SAMPLE CALCULATION

- [ ] **Table 1** is your raw data from Day 1, section 4. **Table 2** is your processed data from Day 2, section 1.
- [ ] Give each table a numbered title *above* it. Every header names the variable and its unit; the cells hold numbers only.
- [ ] Keep every time to the hundredth of a second (`1.40`, not `1.4`).
- [ ] Show one sample calculation for each new column: the average, the ±, and t^2.

EXEMPLAR

Data

Table 1. Time for the cart to reach each mark, released from rest.

POSITION X (M)	TRIAL 1 T (S)	TRIAL 2 T (S)	TRIAL 3 T (S)	AVERAGE T (S)
0.20	1.02	0.95	1.08	1.02
0.40	1.41	1.36	1.45	1.41
0.60	1.72	1.66	1.75	1.71
0.80	1.98	1.93	2.02	1.98
1.00	2.24	2.17	2.21	2.21

Table 2. Processed data. The ± is half the range of the three trials; t^2 uses the average time.

POSITION X (M)	AVERAGE T (S)	± T (S)	T^2 (S^2)
0	0	—	0
0.20	1.02	0.07	1.04
0.40	1.41	0.05	1.99
0.60	1.71	0.05	2.92
0.80	1.98	0.05	3.92
1.00	2.21	0.04	4.88

Sample calculations (the 0.20 m mark):

```
average = (1.02 + 0.95 + 1.08) / 3 = 1.02 s
± = (1.08 − 0.95) / 2 = 0.07 s
t² = (1.02 s)² = 1.04 s²
```

Step 6 · Analysis

TWO GRAPHS, THEN SLOPE → A

- [ ] **Figure 1:** position vs average time, with a smooth curve through the points and the origin (your Day 1 graph).
- [ ] **Figure 2:** position vs t^2, with **one ruled best-fit line**, not connect-the-dots (your Day 2 graph).
- [ ] Both graphs: a numbered title, axes labeled with the variable and unit, and the data shown as dots.
- [ ] Then write four short parts: **slope** (two points read off the line), **acceleration** (a = 2 × slope), **uncertainty** (steepest and shallowest lines), and a **result** line.

EXEMPLAR

Analysis

The position–time graph curves upward (Figure 1), so the cart was speeding up. To test whether the acceleration was constant, we plotted position against time squared (Figure 2).

Figure 1. Position vs time for the cart, released from rest.

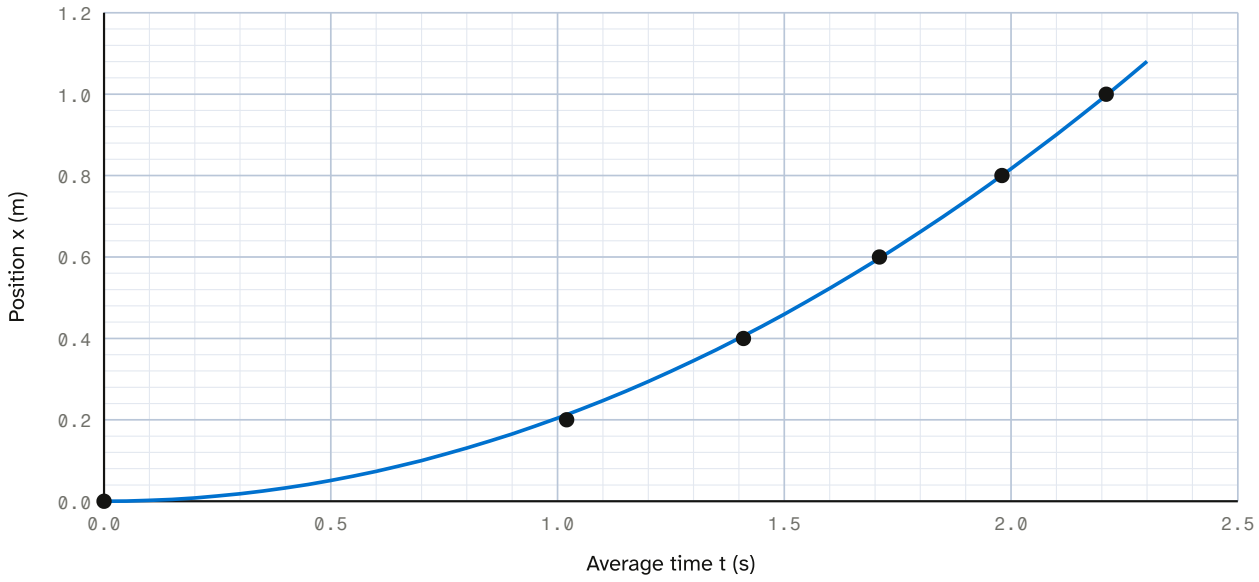
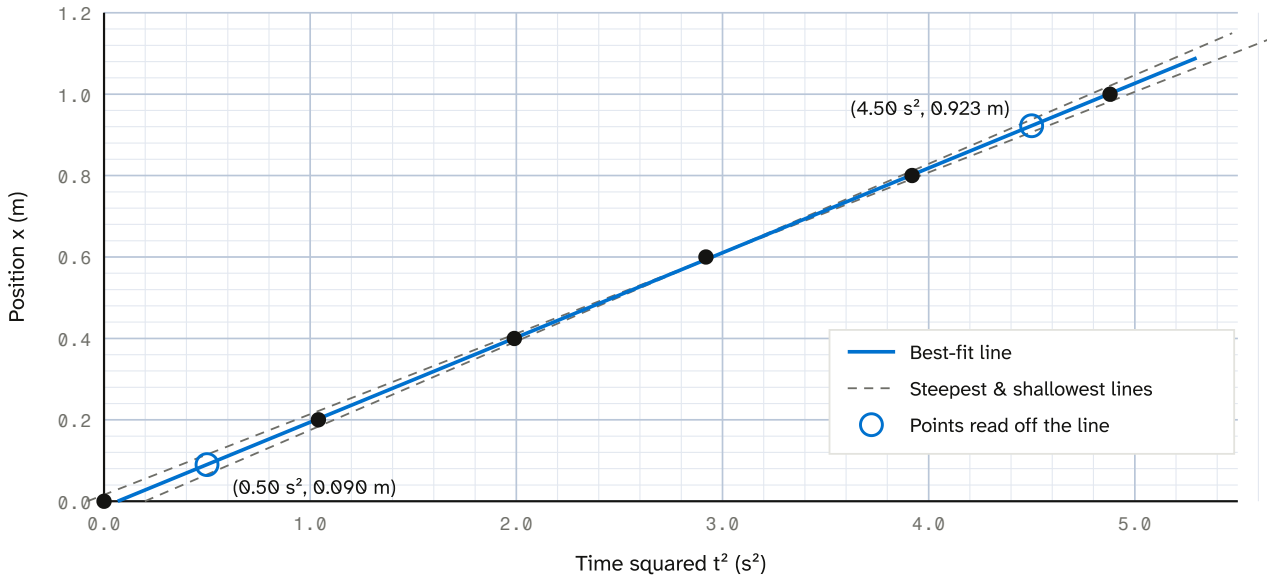


Figure 2. Position vs time squared, with the best-fit line.



EXEMPLAR, CONTINUED

1. Slope. We read two points off the best-fit line, far apart, rather than using data points: (0.50 s², 0.090 m) and (4.50 s², 0.923 m).

```
slope = rise / run = (0.923 m − 0.090 m) / (4.50 s² − 0.50 s²) = 0.833 m / 4.00 s²
= 0.208 m/s²
```

2. Acceleration. Comparing $x = \frac{1}{2}at^2$ with $y = mx$, the slope is ½a, so

```
a = 2 × slope = 2 × 0.208 m/s² = 0.416 m/s²
```

3. Uncertainty. The steepest line that still fits the points has a slope of 0.218 m/s², giving a = 0.436 m/s². The shallowest has a slope of 0.198 m/s², giving a = 0.396 m/s².

```
± = (0.436 − 0.396) / 2 = 0.020 m/s²
```

Result: $a = 0.42 \pm 0.02$ m/s². We rounded 0.416 to 0.42 because the uncertainty is in the hundredths place.

Step 7 · Conclusion

CLAIM · EVIDENCE · REASONING · UNCERTAINTY

- ☐ **Claim:** answer the question in one or two sentences, and give your value of a with its ±.
- ☐ **Evidence:** point to your graphs and numbers, including your time ratio from Day 1, 6(b).
- ☐ **Reasoning:** explain why a straight x–t² line means the acceleration was constant.
- ☐ **Uncertainty:** two sources, and for each say which way it pushes a. Finish with one improvement.

EXEMPLAR

Conclusion

The x–t^2 graph was a straight line, which supports the hypothesis that the cart's acceleration was constant. From the slope, a = 0.42 ± 0.02 m/s^2.

Evidence. The position–time graph curves upward, so the cart covered more distance each second. Reaching 0.80 m took 1.98 s, which is 1.94 times the 1.02 s it took to reach 0.20 m; we predicted 2 times. When position is plotted against t^2, all five points lie close to one straight line.

Reasoning. A straight line through the origin means x is proportional to t^2. That is what $x = \frac{1}{2}at^2$ predicts when a is constant, so the slope equals ½a.

Sources of uncertainty.

- **Reaction time.** The timer has to start the stopwatch on release and stop it at the mark, and each takes a moment. Stopping late makes every time too long, which makes the slope, and so a, too low. Our line crosses the position axis slightly below zero, at about −0.01 m, which fits times that are a little too long.
- **Judging when the cart reaches the mark.** Watching from an angle makes it hard to tell exactly when the front of the cart crosses the tape. This affects the short runs most: the 0.20 m mark had the biggest spread, ±0.07 s, because a small timing error is a large fraction of a 1 s run. This error can push a either way.

Improvements. Film each run with a phone in slow motion and read the times frame by frame, which removes reaction time. A longer ramp would also help, because longer runs make each timing error a smaller fraction of the measured time.

Before you turn it in

CHECK ONLY WHAT IS TRUE

- ☐ All seven sections, in order, with headings
- ☐ Hypothesis uses If … then … because
- ☐ Materials give sizes and tool precision
- ☐ Procedure is numbered and repeatable
- ☐ Tables: numbered titles; headers with units
- ☐ Cells: numbers only; times to 0.01 s
- ☐ One sample calculation per new column
- ☐ Both graphs: titles, axis labels with units
- ☐ x–t: smooth curve; x–t^2: one ruled line
- ☐ Slope from points on the line, with its unit
- ☐ a stated with a ± and a unit
- ☐ Two uncertainty sources, each with a direction
- ☐ My numbers, my words — not the exemplar's